✓ Algebraic Structures

UNIT-4 DSTL

→ non-empty sets $\xrightarrow[\text{with}]{\text{equipped}}$ one or more operations

that ↓ satisfy

specific axioms or laws

→ expressed as a pair (A, F), where

* A is a non-empty set (called the underlying set or carrier)

* F is a collection of operations defined on A (e.g., binary or unary or constant operations)

A non-empty set S is called Algebraic Structure w.r.t. binary operation $*$ if $(a * b) \in S$

$\forall (a * b) \in S$

$\forall a, b \in S$

'$*$' is closure operation on S

Properties of binary operations

i) Closure property

ii) Associative property

iii) Identity property

iv) Inverse property

v) Commutative property

vi) Idempotent

	Properties of Binary Operations				
	Closure	Associative	Identity	Inverse	Commutative
Algebraic Structure	✓	✗	✗	✗	✗
Semigroup	✓	✓	✗	✗	✗
Monoid	✓	✓	✓	✗	✗
Group	✓	✓	✓	✓	✗
Abelian Group	✓	✓	✓	✓	✓

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Algebraic Structure

↳ closure property

↳ Let A be the non-empty set. An operation $*$ is said to be **closed** on A if $\forall a, b \in A$ $a * b \in A$

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e.g.,

$N, +$	Yes	$Z, +$	Yes	$R, +$	Yes
$N, -$	No	$Z, -$	Yes	$R, -$	Yes
$N, \times$	Yes	$Z, \times$	Yes	$R, \times$	Yes
$N, \div$	No	$Z, \div$	No	$R, \div$	No

Semigroups

Let $(A, *)$ be an algebraic structure, where $*$ is a binary operation on A.

$(A, *)$ is called semigroup if it satisfies the following conditions.

(i) Closure

(ii) Associative

$$\boxed{(a * b) * c = a * (b * c)}$$

e.g.,

$N, +$ Yes	$Z, +$ Yes	$R, +$ Yes
$N, \times$ Yes	$Z, -$ No	$R, -$ No
	$Z, \times$ Yes	$R, \times$ Yes

Monoids

Let $(A, *)$ be an algebraic structure, where $*$ is a binary operation on A. $(A, *)$ is called Monoid if the following conditions are satisfied.

(i) Closure property
(ii) Associative property
(iii) Identity

$a, e \in A$

$$\boxed{a * e = e * a = a}$$

e.g.,

$N, +$ No

$Z, +$ Yes

$R, +$ Yes

$N, \times$ Yes

$Z, \times$ Yes

$R, \times$ Yes

Group Let $(A, *)$ be an algebraic structure, where $*$ is a binary operation on A. $(A, *)$ is called Group if the following conditions are satisfied

(i) Closure (ii) Associative

(iii) Identity (iv) Inverse

$$\boxed{a * a^{-1} = e}$$

e.g., $N, \times$ No $\qquad Z, +$ Yes $\qquad R, +$ Yes

$$\boxed{a, a^{-1}, e \in A}$$

$Z, \times$ No $\qquad R, \times$ No

Additive inverse

Multiplicative inverse

$$3 \times \frac{1}{3} = 1$$

$3 \in N \qquad 1 \in N \qquad 1/3 \notin N$

Abelian Group

Let $(A, *)$ be an algebraic structure, where $*$ is a binary operation on A.

$(A, *)$ is called Abelian Group if the following conditions are satisfied

(i) Closure property
(ii) Associative property
(iii) Identity property
(iv) Inverse property
(v) Commutative property

$$\boxed{a * b = b * a}$$

e.g.) $Z, +$ Yes $\qquad R, +$ Yes

Prove that the set Z of all integers with binary operation $*$ defined by $a*b = a+b+1$ such that $\forall\, a,b \in Z$ is an abelian group.

$a*b = a+b+1$

(i) Closure property

$\forall a,b \in Z$

$a*b = a+b+1 \in Z$

(ii) Associative property

$a*(b*c) = (a*b)*c$

$= a+b+c+2$

(iii) Identity property

$a*e = a$

$\cancel{a}+e+1 = \cancel{a}$

$\boxed{e=-1}$

(iv) Inverse property

$a*a^{-1} = e$

$a+a^{-1}+1 = -1$

$a^{-1} = -2-a \in Z$

for any $a \in Z$

Z satisfies _____ property.

(v) Commutative property

$$a * b = b * a$$

$$a+b+1 = b+a+1 \in Z$$

$$2+3+1 = 3+2+1 = 6 \in Z$$

$$-2+3+1 = 3-2+1 = 2 \in Z$$

Z satisfies commutative property

Hence, Z is an Abelian Group.

Q: $G = \{\{1, -1, i, -i\}, \times\}$

$H = \{1, -1\}$

Prove that H is subgroup of G under operation $\times$.

A:

$\times$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

$G = \{\underline{\underline{\{1, -1, i, -i\}}}, \times\}$

(i) Closure

$\forall a, b \in G$

$a * b \in G$

All the values belong to G.

$\therefore$ G satisfies Closure property.

(ii) Associative

$\forall a, b \in G$

$a*(b*c) = (a*b)*c$

$1 \times (-1 \times i) = (1 \times -1) \times i$

$= -i$

$\boxed{G \text{ is a Group}}$

(iii) Identity

$\boxed{a * e = a}$

$e = 1$ always for multiplication

$1 \times 1 = 1 \in G \quad i \times 1 = i \in G$

$-1 \times 1 = -1 \in G \quad -i \times 1 = -i \in G$

(iv) Inverse

$a * a^{-1} = e$

$a \times a^{-1} = 1$

$1 \times 1^{-1} = 1 \quad | \quad i \times i^{-1} = 1$

$-1 \times (-1)^{-1} = 1 \quad | \quad -i \times (-i)^{-1} = 1$

In given question we have to prove that H is subgroup of G or not.

$$H = (\{1, -1\}, \times)$$

(i) Closure property $\quad a \times b \in H$

(ii) Associative property $\quad a \times (b \times c) = (a \times b) \times c$

(iii) Identity property $\quad a \times e = a$

$e \longrightarrow$ identity element

(iv) Inverse property $\quad a \times a^{-1} = e$

Modulo Operation

$+_m \longrightarrow$ modulo addition

$\times_m \longrightarrow$ modulo multiplication

$a +_m b = a+b$ if $a+b < m$

$= a+b-m$ if $a+b \geq m$

OR

$(a+b) \% m$

$a \times_m b = ab$ if $ab < m$

$= ab \% m$ if $ab \geq m$

a	b	$a +_4 b$	$a \times_4 b$
1	2	3	2
2	3	$5-4=1$ OR $5 \% 4 = 1$	$6 \% 4 = 2$

$m = 4$

Modulo Addition & Multiplication Questions

Q: Prove that $G = (\{0,1,2,3\}, +_4)$ is a group.

Q: Prove that $G = (\{0,1,2,3\}, \times_4)$ is a group.

Order of a Group (Cardinality)

- Number of elements in a group G
- Denoted by $|G|$

Cyclic Group

Generator: An element which can generate an entire group

OR

For a group $(G, *)$, an element $a \in G$ is called generator if all elements of G can be represented using power of a.

Example :- $G = (\{0, 1, 2, 3\}, +_4)$

$$G = (\{1^4, 1^1, 1^2, 1^3\}, +_4)$$

$1^1 = 1$

$1^2 = 1+1 = 2$

$1^3 = 1+1+1 = 3$

$1^4 = 1+1+1+1 = 4-4 = 0$

$a +_4 b = a+b$ if $a+b < 4$

$a +_4 b = (a+b) \% 4$ if $a+b \geq 4$

OR

$(a+b) - 4$

$0^1 = 0$

$0^2 = 0+0 = 0$

$0^3 = 0+0+0 = 0$

not a generator

$2^1 = 2$

$2^2 = 2+2 = 4-4 = 0$

$2^3 = 2+2+2 = 6-4 = 2$

$2^4 = 2+2+2+2 = 8 \% 4 = 0$

Problem:

Given a set Q of all rational numbers other than 1 with $\boxed{a * b = a + b - ab}$. Show that $(G, *)$ is a group.

Solⁿ: (I) Closure property

$a, b \in Q$

$a + b - ab \in Q$

Satisfies closure property

(II) Associative property

$(a * b) * c = (a + b - ab) * c$

$= (a + b - ab) + c - (a+b-ab)c$

$= a + b - ab + c - (ac + bc - abc)$

$= a + b - ab + c - ac - bc + abc$

$= a + b + c - ab - ac - bc + abc$

$$a*(b*c) = a*(b+c-bc)$$
$$= a+b+c-bc-a(b+c-bc)$$
$$= a+b+c-bc-ab-ac+abc$$
$$= \boxed{a+b+c-ab-ac-bc+abc}$$

(iii) Identity property

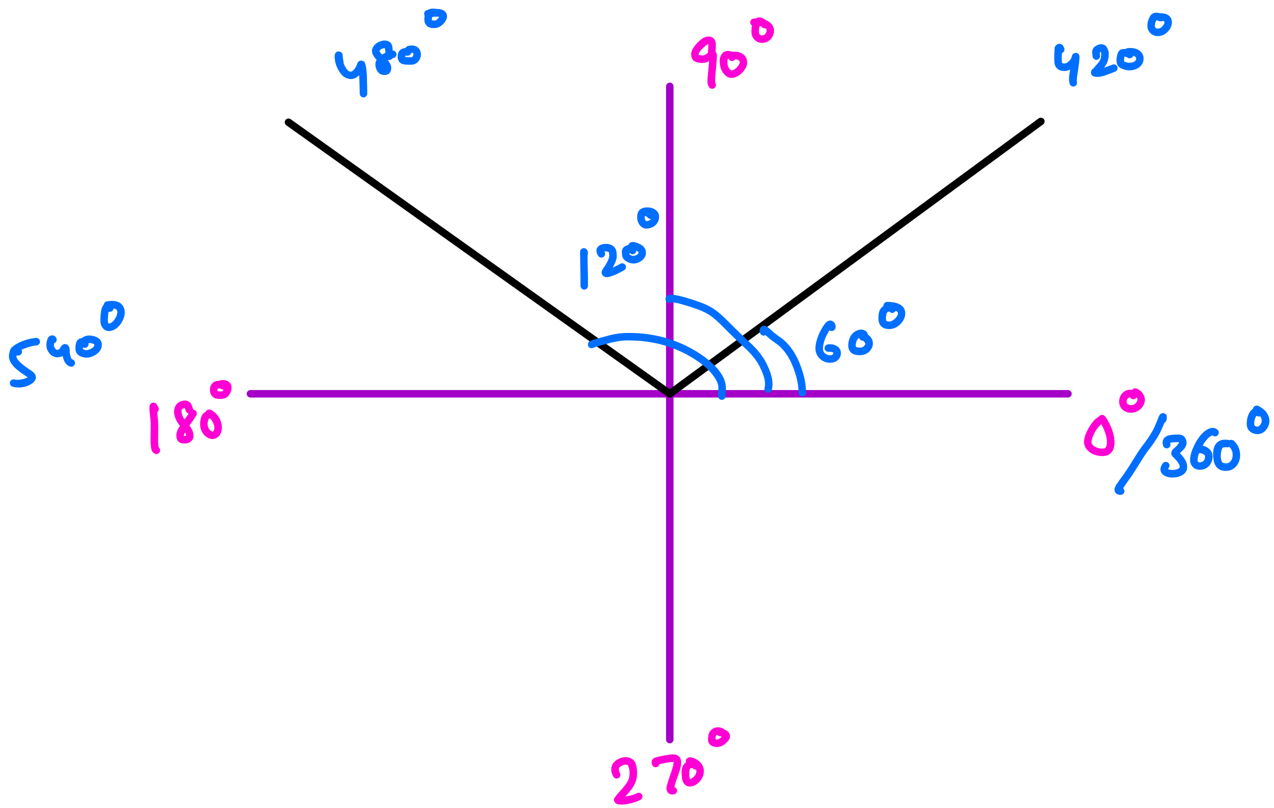
$$a*e = a$$
$$\cancel{a}+e-ae = \cancel{a}$$
$$e(1-a) = 0$$
$$\boxed{e=0}$$

(iv) Inverse property

$$a*a^{-1} = e$$
$$a+a^{-1}-aa^{-1} = 0$$
$$a^{-1}(1-a) = -a$$
$$\boxed{a^{-1} = \frac{-a}{1-a}} \quad a\neq 1,\ a\in Q$$

Q: Let $R = \{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$ and $*$ is a binary operation so that a and b in R, $a * b$ is overall angular rotation corresponding to successive rotations by a and b. Show that $(R, *)$ is a group.

rotations are adding up



*	0°	60°	120°	180°	240°	300°
0°	0°	60°	120°	180°	240°	300°
60°	60°	120°	180°	240°	300°	0°
120°	120°	180°	240°	300°	0°	60°
180°	180°	240°	300°	0°	60°	120°
240°	240°	300°	0°	60°	120°	180°
300°	300°	0°	60°	120°	180°	240°

finite group

$*$	0°	60°	120°	180°	240°	300°
0°	0°	60°	120°	180°	240°	300°
60°	60°	120°	180°	240°	300°	360° 0°
120°	120°	180°	240°	300°	0°	60°
180°	180°	240°	300°	0°	60°	120°
240°	240°	300°	0°	60°	120°	180°
300°	300°	0°	60°	120°	180°	240°

addition $+$

$$120^\circ + 300^\circ$$
$$\downarrow$$
$$420^\circ$$

$$480^\circ$$

(i) Closure Property

All outputs $\in R$.

(ii) Associative Property

$$(0^\circ * 60^\circ) * 120^\circ = 60^\circ * 120^\circ = 180^\circ$$

$$0^\circ * (60^\circ * 120^\circ) = 0^\circ * 180^\circ = 180^\circ$$

(iii) Identity Property

$$a * e = a$$

$$60^\circ * e = 60 \text{ or } 60 + e = 60$$

$$e = 0$$

(iv) Inverse Property

$$a * a^{-1} = e$$

$$a + a^{-1} = e$$

$$60 + a^{-1} = 0$$

$$a^{-1} = -60 = 300^\circ$$

Q: Show that $(G, +_8)$ is an Abelian Group where $G = (0,1,2,3,4,5,6,7)$.

$+_8$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

$$6 +_8 a^{-1} = 0$$

(i) Closure

$\forall a, b \in G$

$a +_8 b \in G$

$\therefore$ satisfies closure property

(ii) Associative

$(a +_8 b) +_8 c$

$= a +_8 (b +_8 c)$

e.g., $(1 +_8 2) +_8 3 = 3 +_8 3$

$= 6$

Q: Show that $(G, +_8)$ is an Abelian Group where $G = (0,1,2,3,4,5,6,7)$.

$+_8$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(iii) Identity

$\boxed{e = 0}$

$a * e = a$

$\cancel{1} +_8 e = \cancel{1}$

$e = 0$

(iv) Inverse

$0 +_8 0 = 0$

$1 +_8 7 = 0$

$2 +_8 6 = 0$

$3 +_8 5 = 0$

$\vdots$

vice-versa

(v) Commutative

$a * b = b * a$

$1 +_8 2 = 3$

$2 +_8 1 = 3$

Q: Prove that the set $G = (1,2,3,4,5,6)$ is a finite Abelian group of order 6 w.r.t. multiplication modulo 7.

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

(i) Closure

$\forall a, b \in G$

$a \times_7 b \in G$

(ii) Associative

$(2 \times_7 3) \times_7 6 =$

$2 \times_7 (3 \times_7 6) =$

(iii) Identity

$a * e = a$

$\cancel{2} \times_7 e = \cancel{2}$

$\boxed{e = 1}$

(iv) Inverse

$a * a^{-1} = e$

$2 \times_7 a^{-1} = 1$

$(1,1)\,(2,4)\,(3,5)\,(6,6)$

$(4,2)\,(5,3)$

(v) Commutative

$2 \times_7 3 = 6$

$3 \times_7 2 = 6$

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Cyclic Group

A group G is said to be cyclic if for some a in G, every element x in G can be expressed as a^n, for some integer n.

Thus G is generated by a i.e. $G = (a)$ or $[a]$

e.g., $A = \{1, 2, 3, 4, 5, 6\}$ $\times_7$

$3^1 = 3$

$3^2 = 3 \times_7 3 = 2$

$3^3 = 3^2 \times_7 3 = 2 \times_7 3 = 6$

$3^4 = 3^3 \times_7 3 = 6 \times_7 3 = 4$

$3^5 = 3^4 \times_7 3 = 4 \times_7 3 = 5$

$3^6 = 3^5 \times_7 3 = 5 \times_7 3 = 1$

$\boxed{3 = \text{Generator}}$

Subgroups

Let $(A, *)$ be a group and B be a subset of A, $(B, *)$ is said to be a subgroup of A if $(B, *)$ is also a group by itself. Suppose we want to check whether $(B, *)$ is a subgroup for a given subset B of A.

$$B \subseteq A$$

if $G\,(A, *)$ be the group
$H\,(B, *)$ be the subgroup

then $H \leq G$

H is subgroup of G

Normal Subgroup

$H \leq G$, if $\underset{\text{Left coset}}{xH} = \underset{\text{Right coset}}{Hx} \quad \forall x \in G$

H is Normal Subgroup of G.

If every left coset of H is equal to right coset.

Coset

Let $(G, *)$ be a group and let H be a subgroup of G. For $a, b \in G$ we say a is congruent to b modulo H, written as a equiv b mod H, if $a * b^{-1} \in H$.

The congruence relation is an equivalence relation on G.

$(G, +)$, $(H, +)$

left coset

right coset

$a \in G$

$H \leq G$

Not always

$a + H = H + a$

Q: Let $G = (\mathbb{Z}, +) = \{\ldots -3, -2, -1, 0, 1, 2, 3, \ldots\}$ be a Group.

$H = (3\mathbb{Z}, +) = \{\ldots -6, -3, 0, 3, 6, \ldots\}$ be its Subgroup.

Find all Right cosets of H in G. $(H+a), a \in G$

A:

$H+0 = \{\ldots -6, -3, 0, 3, 6, \ldots\}$

$H+1 = \{\ldots -5, -2, 1, 4, 7, \ldots\}$

$H+2 = \{\ldots -4, -1, 2, 5, 8, \ldots\}$

$H+3 = \{\ldots -3, 0, 3, 6, \ldots\}$

$H+4 = \{\ldots -2, 1, 4, 7, \ldots\}$

$H+5 = \{\ldots -1, 2, 5, 8, \ldots\}$

Similarly $H+6 = H$, $H+7 = H+1$, $H+8 = H+2$

No. of right cosets = 3 $\quad H, H+1, H+2$

$$G = (H) \cup (H+1) \cup (H+2)$$

Properties

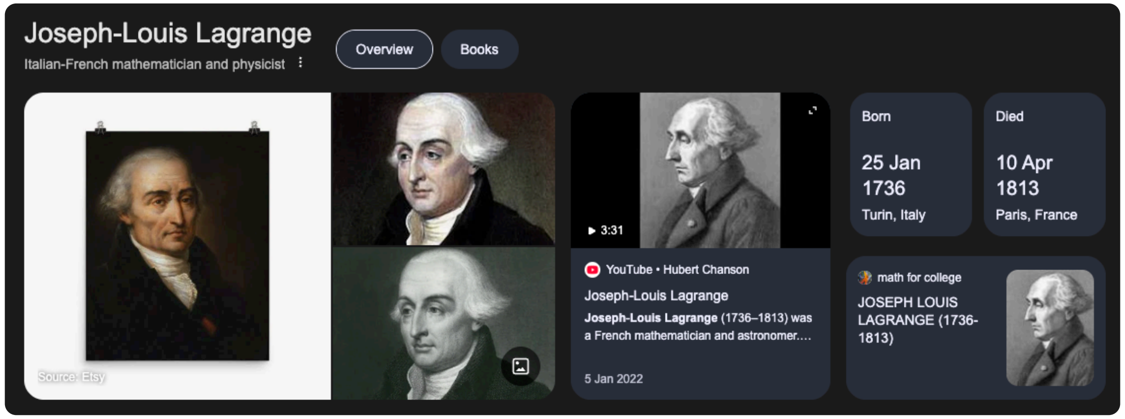
① Any two right cosets (or left) are either Disjoint or identical.

② Group G is union of all distinct right cosets of H in G.

If H is subgroup of G, then

$aH \rightarrow$ left coset
$Ha \rightarrow$ right

1. $Ha = H$ iff $a \in H$
2. $Ha = Hb$ iff $ab \in H$
3. Any two Right cosets are either Disjoint or Identical.
4. Set G is union of all Right Cosets of H in G.



Lagrange's Theorem

If G is a finite group and H is a subgroup of G, then

(a) $|H|$ divides $|G|$

(b) The number of distinct left (right) cosets of H in G is $|G|/|H|$.

The order of each subgroup of a finite group is divisor of the order of the Group.

[divisible]

$o(G)/o(H)$

Let $o(G)=n$, $o(H)=m$

m/n is to be proved

$H=\{h_1, h_2, h_3, \dots, h_m\}$

$n = km$

$a_1 \in G$

$Ha_1 = \{h_1 a_1, h_2 a_1, h_3 a_1 \dots h_m a_1\}$

$a_2 \in G$

$Ha_2 = \{h_1 a_2, h_2 a_2, h_3 a_2, \dots h_m a_2\}$

$\vdots$ upto n cosets

but NO REPEATED COSETS

k distinct cosets

$o(G) = 6$

order of subgroup of this group G
will always divide the order of the
group G.